Gluconeogenesis in Cattle: Significance and Methodology[1]

J. W. YOUNG
Department of Animal Science
Iowa State University
Ames 50011

ABSTRACT

Gluconeogenesis is a continual process that is of great importance in ruminants because almost all dietary carbohydrates are fermented to volatile fatty acids in the rumen. In turn, propionate is the only major volatile fatty acid that contributes to gluconeogenesis. Many different techniques and analytical procedures are involved in studying ruminant gluconeogenesis. Glucose kinetics can be examined by single-injection or continuous-infusion isotope dilution techniques with a variety of glucose labels. Correcting for recycling of label is an important consideration. Absorption of glucose from the gut can be measured by combining arterial-venous differences and flow rates of portal blood or can be estimated by determining the amount of glucose plus α-glucose polymers passing into the small intestine. Production of propionate in the ruminoreticulum can be measured by isotope dilution techniques. Quantitating the conversion of propionate to glucose requires the use of [carbon-14] propionate with careful corrections for propionate carbon entering the citric acid cycle before incorporation into glucose. The same fundamental techniques used with propionate are required to quantitate the contributions of amino acids and other precursors to glucose. In vitro studies of gluconeogenic enzymes, and cellular, tissue, or organ preparations provide valuable insights into the gluconeogenic processes and controls but must be validated by in vivo experiments. Progress has been considerable in understanding some aspects of ruminant gluconeogenesis, but many more studies will be required to obtain a complete understanding.

SIGNIFICANCE OF GLUCONEOGENESIS

In ruminants, gluconeogenesis is of great importance at all times and is most important after feeding when there is a large influx of metabolites into the bloodstream. In nonruminant species, the rate of gluconeogenesis is lowest after feeding when glucose is being absorbed and is highest during fasting when no exogenous glucose is provided. In ruminants, dietary carbohydrates are fermented to short-chain volatile fatty acids in the rumen and often less than 10% of the glucose requirement is absorbed from the ruminant digestive tract (2, 6, 67). Thus, gluconeogenesis must provide up to 90% of the necessary glucose in ruminants. Carbohydrate digestion and glucose absorption can be varied or influenced by many factors (3, 86).

Meeting glucose needs can be a tremendous metabolic challenge for a high-producing dairy cow. At peak lactation, Beecher Arlinda Ellen produced 89 kg of milk daily (Successful Farming, Page A6, Jan., 1976). If the milk was 4.9% lactose, she secreted 4.43 kg of lactose daily. If, as Bickerstaffe et al. (15) found with cows and Bergman and Hogue (9) found with sheep, about 60% of the glucose entry is used for lactose, this cow required about 7.4 kg of total glucose daily. If she had to synthesize 90%, or 6.6 kg of this glucose, this is a major metabolic undertaking. We have data indicating that 160- to 205-kg steers fed slightly above maintenance will need to synthesize about .6 kg of glucose daily (67, 105). Hence, even in nonlactating ruminants, the gluconeogenic "machinery" must be operating continually at a rapid rate, and any "breakdown" can result in serious metabolic disorders.

There likely are major differences between ruminants and nonruminants in the gluconeo-

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genic process and its control (4, 55). At least there are differences in the degree the process is turned on. The ruminant liver releases glucose during both the fed and fasted states, with the greatest production occurring after feeding (4). The liver of nonruminants in the fed state has a net uptake of glucose, but during fasting there is a net release of glucose. Thus, ruminant gluconeogenesis is normally a process of both continual importance and considerable intensity. Another difference is that both gluconeogenic (55) and lipogenic rates (69) are elevated when feed intake is abundant in ruminants whereas only lipogenesis (104) is elevated in the nonruminant when intake is abundant. Nonruminant gluconeogenesis is greater during a period of fasting than during normal intake. Thus, there may be differing physiological and biochemical controls linking gluconeogenesis and lipogenesis for the two types of animals. The literature on gluconeogenesis and glucose metabolism in ruminants (mostly with sheep) was reviewed and summarized recently by both Bergman (7) and Leng (55).

There are a number of practical applications of research on gluconeogenesis. First, both bovine ketosis and ovine pregnancy toxemia seem to be related to a shortage of glucose or gluconeogenic precursors. Second, milk fat depression, which occurs when cows are fed a high proportion of grain, seems to be related to an overabundance of glucose or propionate for metabolism in these cows. Third, it is possible that milk production in some cows may be limited by the supply of glucose (2) or by a glucogenic capacity inadequate to support maximum milk production. Fourth, certain feed additives that promote growth or increased feed efficiency may act through changes in glucose metabolism. For example, monensin (26, 66), a new feed additive for cattle, increases propionate production which may in turn increase gluconeogenesis.

METHODOLOGY RELATED TO GLUCONEOGENESIS

This manuscript will attempt to include representative papers, key ideas, and various viewpoints related to methodology for ruminant gluconeogenesis rather than including all references relating to individual techniques. The manuscript also is influenced by experiences in the author's laboratory. A few years ago, we began an effort to quantitate glucose kinetics in cattle under specified conditions and then to quantitate the various precursors contributing to the total glucose metabolism (67, 105). Thus, this manuscript will deal largely with various aspects of in vivo glucose kinetics in ruminants.

Methods and Problems in Glucose Kinetic Studies

There are five fundamental questions in glucose kinetic studies; first, method of isotope administration; second, isotope label and molecular positioning of the label of glucose; third, feeding regime; fourth, sampling interval; and fifth, analysis to determine specific activity of glucose.

In relation to the first consideration, there are three basic methods of isotope administration: single injection, continuous infusion, and primed continuous infusion. All three have been used, but the single injection and the primed continuous infusion method have been used most. There has been a question as to whether these two methods give the same results; i.e., whether they measure the same aspects of glucose metabolism. Shipley and Clark (77) and White et al. (96) with cattle obtained values for irreversible loss of glucose that were not significantly different for the two methods. A number of other workers, however, have not obtained similar results. Steele et al. (80) recommended primed continuous infusions because they felt that single injections of $[^{14}C]$glucose gave incorrect results because of slowly equilibrating glucose pools within the body. In a recent paper, however, Hetenyi and Norwich (40) reviewed the mathematical basis for calculations from both single injections and continuous infusion and concluded that the two methods will yield the same value for metabolic turnover rate if the data are analyzed properly.

We have used single injections followed immediately by primed continuous infusions to compare the two methods (105) and found no significant difference in irreversible loss. The primed continuous infusion allows evaluation of whether the animal was in steady state during the experiment, but the only aspect of glucose kinetics that can be measured accurately is irreversible loss. Unidirectional fluxes and end products of glucose metabolism, however, can be measured best with continuous infusion. The single injection technique does not allow evaluation of whether steady state is achieved, but it does allow calculation of pool size, total entry rate, irreversible loss, and recycling. Thus, more information can be obtained from single injections.

An important consideration in analyzing single-injection kinetic data is how many "components" or compartments characterize the metabolite. Steele and coworkers (78, 79) have examined various aspects of compartments or pools in dogs. Shipley and Clark (77) have discussed the difficulties in distinguishing the number of components in an in vivo system. White et al. (96) published results from single injections of labeled glucose into sheep, and their kinetic analyses varied between two and three components. We have concentrated on obtaining adequate quantity and quality of data (105) to evaluate a two-component model because two major components from single injection kinetics give irreversible loss values averaging almost the same as irreversible losses from continuous infusions. Judson and Leng (45) show three definite components in lactating cows.

The second consideration, relating to type of labeled glucose, includes glucose uniformly labeled with ^{14}C, glucose labeled in specific carbons with ^{14}C, and glucose labeled in specific positions with ^{3}H. There is no easy and straightforward answer to this question, although uniformly labeled [^{14}C] glucose probably has been the most widely used glucose tracer. One of the problems, however, is chemical recycling of labeled carbon through three-carbon compounds and CO_2 returning to glucose.

To avoid some aspects of the chemical recycling problem, Katz and Dunn (49) used [2-^{3}H] glucose and Dunn et al. (27) used [6-^{3}H] glucose. An ideally labeled tracer for estimating irreversible loss is one with the label lost at an early stage of metabolism. In [2-^{3}H] glucose, the tritium would be lost as water at the hexose phosphate isomerization reaction (23). Katz and Dunn (49) used [2-^{3}H] glucose mixed with [U-^{14}C] glucose and found that, in rats, the half-life of tritium-labeled glucose was 22 min and that of [^{14}C] labeled glucose was 34 min. The [2-^{3}H] glucose corrects for recycling of glucose via glycogen and for recycling of glucose carbon via the Cori cycle (resynthesis of glucose from lactate). Glucose turnover calculated from tritium data was 3% of the glucose pool/min, as compared with 2% from ^{14}C data. Thus, up to half of the ^{14}C of the catabolized glucose was recycled. Altszuler et al. (1) compared [2-^{3}H] glucose, [3-^{3}H] glucose and [6-^{14}C] glucose. The [6-^{14}C] glucose has been used to correct for recycling; however, values with [3-^{3}H] glucose were similar to those with [6-^{14}C] glucose. Glucose turnovers using [2-^{3}H] glucose were 1.5 times greater than from [6-^{14}C] glucose.

Because the loss of tritium from position 6 of glucose occurs mainly in the metabolism of pyruvate, Dunn et al. (27) used [6-^{3}H-^{14}C] glucose to provide an estimation of Cori cycle activity in rats. There would be no tritium loss when glucose went directly to glycogen, so they observed complete retention of tritium in muscle glycogen and partial loss of tritium in liver glycogen. The half-times of [^{3}H] glucose and [^{14}C] glucose were 32 and 41 min. The fractional turnover rate as measured with tritium was 2.2% of the glucose pool/min in contrast to 1.7%/min for ^{14}C. They proposed that [6-^{3}H] glucose in turnover studies corrects for recycling and gives an accurate measure of new glucose production. One hydrogen from the methyl group of pyruvate is lost in the carboxylation step. If oxalacetate equilibrates with malate and fumarate, hydrogen originating in the methyl group of pyruvate would be exchanged with protons.

We have used [6-^{3}H] glucose in cattle to eliminate Cori cycle recycling of label. Ruminants seem to differ from other mammals and chemical recycling is minimal, 4 to 5%, if [6-^{3}H] glucose is used (8, 20). There is some confusion as to types of recycling. For example, [6-^{3}H] glucose eliminates Cori cycle recycling, but there is still a large component of recycling, possibly from locational pools in the body when kinetic parameters are calculated by the method of White et al. (96). Hetenyi and Norwich (40) refer to recirculation of label for chemical cycling such as the Cori cycle and use reflux or recycle to refer to the return of tracer to the sampled compartment or volume after a sojourn in some other parts of the system. Judson and Leng (45) compared various glucose labels for estimating total entry rate and resynthesis of glucose in sheep and concluded that [2-^{3}H] glucose and [U-^{14}C] glucose are the most useful for studies of glucose metabolism, particularly for estimating the extent of glucose resynthesis. I feel that [6-^{3}H] glucose is the best to use for most purposes. This seems to be the unofficial consensus of the Tracer Methodology Group (8).

The third consideration, type of feeding regime, is important if one wants accurate results, particularly from a single injection experiment. A number of investigators (10, 13, 31, 45, 105) have begun to feed experimental animals either hourly or every 2 h. This should allow ruminants to be in steady state during experiments. Our results and results of others with primed continuous infusions of tracer indicate that the animals are in steady state. If cattle are fed only twice per day, one would expect an increased absorption of nutrients between 1 and 3 h past feeding (81). This influx of gluconeogenic substrate probably would be detectable as a decrease specific activity of blood glucose in animals where primed continuous infusions were already underway. There would be no way of detecting the change in specific activity after a single injection of isotope. Studies are not available where total 24 h glucose utilization or entry rate is compared for hourly versus 12 h feeding intervals. Gray et al. (38) fed at 1, 2, and 12 h intervals and found that average 24 h propionate production rates were not significantly different. Their results could be used to suggest that total 24-h glucose-entry rate measured by continuous infusion would not be different with various feeding intervals. Single injection measurements should not be used to estimate 24-h glucose-entry rates when the feeding interval is 12 or 24 h.

Experiments in which a single injection of isotopic glucose was given 1 h after feeding and the declining specific activity in blood was followed for almost 7 h have been reported (32, 33). These workers suggest that a single component will explain fully the decreasing slope of specific activity. They report that decreasing specific activity is correlated highly with increasing time after injection such that the coefficients of determination (r^2) for linear regression are $>.95$ in 56 of 58 experiments. One cannot but wonder, however, what effect increased metabolite absorption from the feeding had upon the changing specific activity. It would be helpful to know the relative contributions of various precursors at different times after feeding. Thomson et al. (85) showed that changes in glucose flux caused rapid changes in specific activity of blood glucose during infusions initiated when cows were at a steady state. One also must acknowledge that feeding equal amounts at 1 or 2 h intervals does not correspond to natural patterns of feed consumption in cattle (71).

The fourth consideration, when should sampling be done, has no absolute answer; however, it can be dealt with in generalities. The general belief in our laboratory is that most workers have tended to take fewer samples than are required for best results. Leng (55) also expressed the same concern. In our single-injection experiments, we usually take 20 blood samples from 5 to 240 min postinjection, with the sampling interval being 5 min at the start and increasing up to 20-min intervals at the end (105). With primed continuous infusions, we do not begin sampling until 1 h after the infusion was begun and then sample at 20-min intervals for 3 h, giving a total of 12 blood samples (105). These sampling intervals have allowed us to have a good representation of the specific-activity curves of the injected glucose and to minimize variation caused by one or two data points that may be suspect.

The fifth and final consideration relates to the isolation of glucose for determinations of specific activity. Specific activity of glucose has been determined by preparing potassium gluconate (13, 100), by preparing glucosazone derivative (18, 28), by preparing glucose pentaacetate (43), and by removing contaminating ionic molecules by ion exchange (75, 85). The pentaacetate method is accurate and easier than other derivative methods but is time consuming. Consequently, we have used an ion exchange technique to remove contaminating radioactive organic anions and have shown that this technique gives kinetic results similar to the pentaacetate method (75). This procedure requires much less time per sample than do the derivative methods.

Absorption of Glucose From the Gut

There are two basic ways to estimate glucose absorption. First, by measuring portal vein blood flow rates and atrteriovenous differences, quantities absorbed can be estimated directly (87, 88). Arteriovenous differences and blood flow values measure only net quantities absorbed, however, because one has no idea of the unidirectional removal from blood. Further, estimates from this method would be low because of gut utilization. Second, the maximum amount of glucose available for absorption can be estimated by continually measuring duodenal digesta flow, sampling digesta, and assaying for total α-glucose polymers plus any free glucose. Using both duodenal and ileal re-entrant cannulas could improve the estimate but would not account for possible microbial fermentation in the small intestine.

The first method, using arteriovenous differences, has been used by Katz and Bergman (47, 48), Carr and Jacobson (22), and Wangsness and McGilliard (87, 88). Nutrient absorption from the gut is calculated directly from the product of differences in portal-arterial nutrient concentration and portal blood flow. Katz and Bergman (47, 48) used an indicator-dilution method and simultaneously measured portal and hepatic venous blood flow in sheep during a constant infusion of para-aminohippuric acid into a mesenteric vein. They found an uptake of glucose by the portal bed in all sheep, with portal minus arterial differences ranging from an average of $-1.1 \pm .3$ to $-1.9 \pm .4$ mg/100 ml blood. The negative values reflect utilization of glucose by intestinal cells.

Wangsness and McGilliard used indocyanine green in a dye-dilution method of measuring portal blood in calves (88). The technique was evaluated for measuring absorption by administering [^{14}C]glucose (.35, .70, or 1.40 g glucose/kg body weight) into the duodenum, measuring unabsorbed radioactivity at the terminal ileum, and measuring absorption by multiplying portal-arterial differences in radioactivity times portal blood flow (87). Recovery of the ^{14}C dose averaged 104% for seven experiments and ranged from 89 to 114%. From 36 to 99% of the glucose administered was absorbed as indicated by total blood radioactivity. In such studies it is important to identify the ^{14}C actually in blood glucose because of possible fermentation in the small intestine. Wangsness and McGilliard indicate they made such a distinction, but they did not report the results in their paper (87). It was suggested that accurate blood-flow measurements are most critical for quantitating absorption.

Carr and Jacobson (22) used a method based upon the Doppler shift principle to measure portal blood flow in calves consuming whole milk. In seven calves weighing from 45 to 120 kg, they obtained a mean portal blood flow rate of 40.9 ml/min × kg, and the appearance of total sugars in portal blood accounted for 60% of the lactose intake. The Doppler shift method has the advantage that continuous monitoring may be done with the animal unrestrained and unattended. A disadvantage is that it is necessary to sacrifice animals after a series of experiments to obtain direct calibration of flow. Also, the portal vein is contractile and may change diameter sharply within a given time during an experiment.

The second method, based upon the α-glucose polymer content of ingesta continuously sampled upon entering the duodenum, has been used in cattle by Kentucky workers (46, 59), by Coombe and Smith (24), and by our laboratory (67). Thompson and Lamming (83) also used the same technique in sheep. Starch flow into the small intestine varies widely depending upon intake, rumen fluid turnover, and whether the starch is raw or processed. Also, both the cannulation and indicator methods of determining ingesta flow rates have numerous limitations which were reviewed by MacRae (60) and Faichney (35).

Karr et al. (46) equipped 360-kg Angus steers with re-entrant duodenal cannuals and fed diets containing from 19 to 63% starch. They found that starch passing into the abomasum increased with increasing starch intake, and 16 to 38% of the starch intake passed to the duodenum. Their results indicated that the efficiency of starch digestion in the rumen decreased as the daily starch intake exceeded 2 kg and that a considerable amount of starch can be digested in the small intestine. In that same laboratory, Little et al. (59) cannulated the abomasum and posterior ileum of Angus steers and estimated digestion of starch posterior to the rumen after 200, 400, and 600 g of corn starch were infused into the abomasum twice daily. As more starch was infused, a greater percentage of that infused was recovered in the posterior ileum, indicating reduced digestibility. The starch infused was equivalent to amounts passing from the rumen in earlier experiments (46) and indicates that under these conditions digestion of starch in the intestine may be inadequate for optimum utilization. From a 600-g infusion, about 200 g of starch appeared in the feces.

Coombe and Smith (24) gave preruminant calves, which were equipped with re-entrant cannulas in the small intestine, liquid feeds containing starches. Little starch left the abomasum until 5 h after feeding because much of the starch was associated with the casein clot. About 60% of the starch intake was digested during passage through the small intestine, but the authors did not give figures for undigested starch excreted in feces. Thompson and Lamming (83) used sheep with re-entrant cannulas in the proximal duodenum and found that more starch (up to 20% of dietary) from ground corn entered the duodenum daily when the diet contained long or chopped straw than when it contained ground straw.

In our laboratory (67), the amount of α-glucose polymers (starch) entering the duodenum of re-entrant cannulated steers was measured after steers received either 275 or 500 g of a high-grain diet every 2 h. Glucose in duodenal digesta, as a percentage of that ingested, was 3.6 and 5.2%, and total glucose recovered in duodenal digesta per 8-h collection averaged 20 and 40 g (2.5 and 5 g/h) during low and high feed consumption. Directly measured glucose absorption from such 150-kg steers with a blood flow of 40 ml/min × kg and absorbing 2.5 g of glucose/h would give a portal-arterial difference of .7 mg glucose per 100 ml blood, even if there was no utilization of glucose by the gastrointestinal tissues. Such a small difference would be difficult to detect and impossible to measure accurately by blood flow times difference in concentration.

Production of Propionate in the Rumen

Two basic procedures are available for determining the rumen production of a volatile fatty acid (acetate, propionate, or butyrate). These are the continuous-infusion and the single-injection methods of administering labeled acids. The continuous-infusion method has been used by numerous laboratories (12, 30, 38, 56, 57, 58, 95). The continuous-infusion method has the advantage that interconversions of various volatile fatty acids (VFA) can be measured and results corrected accordingly (12, 30, 56, 57). In determining propionate production rates, however, there is little equilibration of propionate with acetate and butyrate (12). Bergman et al. (12) found that the mean net production rate of propionic acid was only 14% lower than its gross rate, but the mean net rates for acetic acid and butyric acid were 21 and 35% lower than their respective gross rates. Esdale et al. (30) observed little propionate ^{14}C equilibrating with acetate or butyrate and $[^{14}C]$propionate reached constant specific activity after 6 h of infusion.

The single-injection method has been used by Davis (25) for acetate, by Knox et al. (52) for acetate, propionate, and butyrate, by Bauman et al. (5) and Yost (102) for propionate. The single-injection method has the advantage that pool size, turnover rate, and exchange rate can be determined whereas in continuous infusions, only irreversible loss can be determined.

An operational problem with both procedures is that of mixing rumen contents to assure reliable measurements. In the continuous-infusion method, continuous mixing is essential. Wiltrout and Satter (98) utilized hand mixing of rumen contents every 5 min in rumen-fistulated cows given continuous infusions. In single injection experiments, it is important to get thorough mixing when the isotopic tracer is given and just before each rumen sampling.

Another method that sometimes has been used to estimate total in vivo volatile fatty acid production is the in vitro production of volatile fatty acids as measured by concentration changes in a sample of rumen contents taken just after feeding (31, 34, 97). This method would not be valid if fermentation is changed by withdrawal of the sample from the rumen or if VFA accumulation results in end-product inhibition. Thus, the first two methods are the methods of choice for determining volatile fatty acid production in the rumen.

A major problem in measuring propionate production rates is the determination of propionate specific activity. The methods available have been long and laborious. Some have used a silicic acid column and butanol in hexane for separation (56, 57, 58); others (30) have used the celite column of Wiseman and Irwin (99), and some (5, 102) have used the silicic acid column of Ramsey (70). High-pressure liquid chromatography techniques, which are time-saving, more convenient, and more accurate should be explored in the future.

The repeatability and precision of propionate production rates are also important. Gray et al. (38) fed sheep 1000 g of fodder daily with 1-, 2-, or 12-h intervals between feedings. They found that volatile fatty acid production varied considerably on successive days, but total daily production was the same at the three feeding intervals. Leng et al. (56) used a continuous infusion to measure acetate, propionate, and butyrate production rates in grazing sheep. They derived regression equations relating entry rates to rumen concentrations and found that equations from penned and grazing sheep did not differ significantly. Weller et al. (95) used a constant-infusion, isotope-dilution technique known to be satisfactory for VFA production in steady state and tested its applicability where the VFA pool in the rumen could vary irregularly. The method gave mean production rates similar to those when pool size was virtually constant. After being tested in a model system, the method was accepted as suitable for measuring VFA production in grazing sheep.

Studies designed to measure repeatability of propionate production rates in steers were completed recently in our laboratory (102). When 275 g of a high-grain diet were fed every 2 h, the propionate production rate averaged 579 g/day, and the range of values was from 347 to 818; when 500 g were fed, corresponding values were 1032 with a range from 729 to 1447. The average increase in production rate paralleled the average increase in feed intake; thus, it seems that propionate production is proportional to feed intake within a given diet. There was more variation in individual propionate production rates than we had expected, however.

Conversion of Propionate to Glucose

Most studies of propionate metabolism or the conversion of propionate to glucose involve the use of ^{14}C-labeled propionate. One of the first quantitative studies on the conversion of propionate to glucose was that of Bergman et al. (13), who did a continuous infusion of [2-^{14}C] propionate into a rumen vein of conscious sheep. They found that about 50% of the propionate absorbed from the rumen was converted to glucose and 40% was synthesized to other compounds. One-fourth was oxidized to CO_2, but most of this occurred after conversion to glucose. In normal animals, about 27% of the glucose turnover originated from propionate. In this study, no attempt was made to evaluate the "crossing-over" of label into intermediates of the citric acid cycle.

Later studies were conducted by Leng et al. (58) and by Judson et al. (44). Leng et al. (58) infused [2-^{14}C] propionate continuously into the rumen of sheep and observed a plateau specific activity of rumen propionate between 4 and 9 h and of plasma glucose between 6 and 9 h after infusions began. Results indicated that 54% of the glucose synthesized arose from propionate carbon. They suggested that a considerable amount of the propionate converted into glucose was first converted into lactate and suggested that the conversion took place in rumen epithelium.

Although there is a definite conversion of propionate to lactate by rumen mucosa in vitro, Weigand et al. (93) with calves, and Weekes (90) with ewes, challenged the idea that this conversion is a major in vivo reaction. Weigand et al. (93) removed rumen contents from rumen-fistulated steers, replaced the contents with a mixture of volatile fatty acids containing labeled propionate, and concluded that the true conversion of propionate to lactate by the rumen wall averaged 2.3% (range 1.0 to 4.6%). Weekes (90) removed rumen papillae and mucosa from ewes at slaughter and measured amounts of lactate formed from propionate per unit dry matter of papillae. Assuming the in vitro rates represent in vivo rates and using an estimated propionate production rate, the calculated conversion of propionate absorbed from the rumen into lactate and pyruvate averaged 3.0%. Weekes and Webster (91) recently published in vivo results which strongly support their in vitro results and agree with the in vivo results of Weigand et al. (93). Thus, the results of Weigand et al. (93), Weekes (90), and Weekes and Webster (91) agree and make it highly unlikely that there is a major in vivo conversion of propionate to lactate by rumen mucosa during propionate absorption. Their studies, however, do not eliminate a possible conversion of propionate to lactate by the liver or other organs.

Studies that have not considered "crossing over" of label into citric acid cycle intermediates likely have underestimated the glucose formed from propionate (17, 53). A first estimate of the contribution of propionate to glucose synthesis is obtained by dividing the carbon specific activity of blood glucose by the carbon specific activity of the precursor propionate. This gives the transfer quotient, which indicates the fraction of glucose carbon derived from the labeled carbon of propionate (98) but reveals nothing of the nature of intervening reactions. A better index of gluconeogenesis is the extent to which propionate charges or replenishes citric acid cycle intermediates that are removed continually for gluconeogenesis. In the experiment of Wiltrout and Satter (98) where [2-^{14}C] propionate was the precursor and glucose the product, the transfer quotient would be an accurate measure of the contribution of propionate to glucose synthesis only if the intermediates do not mix with other four-carbon citric acid cycle intermediates. This was not the case because there was significant labeling in carbons 3 and 4 of the glucose. If oxaloacetate went to glucose without entering the Krebs cycle, there would be no label in carbons 3 and 4. On the basis of data from degraded glucose, Wiltrout and Satter estimated that 47% of the oxaloacetate formed is used for glucose synthesis and that 53% enters the citric acid cycle. By using these figures with the theoretical distribution of label in glucose formed from [2-^{14}C] propionate, they estimated that 26% of the starting label was lost when propionate was converted to glucose after oxaloacetate entered the citric acid cycle. Their transfer quotients then were multiplied by 1.35 to obtain a value of 61% for the fraction of glucose contributed by propionate. The values were not corrected for the incorporation of $^{14}CO_2$ into metabolic intermediates, but this source of potential error is probably smaller than the analytical errors in other phases of the work.

Black and his colleagues have been the most persistent in warning of the dangers of not considering "crossover." One of the most thorough discussions of "crossover" and how to correct for it was presented in the Ph.D. dissertation of Thompson (84) from Black's laboratory. The study was designed to estimate the extent of "crossover" and develop ways to correct for it. Transfer quotient is defined as the percentage of product derived from the radioactive substrate. Transfer quotient deals specifically with radioactivity transfer, and it is possible to misinterpret the significance of the transfer quotient and to encounter an error of some magnitude in assessing the percentage of the product carbon derived from substrate carbon. The misinterpretation occurs with substrates labeled in one specific position when there is crossover in the citric acid cycle. Appropriate corrections must be made.

The corrections are made after finding the fraction of oxaloacetate going to phosphoenolpyruvate and then on to glucose synthesis (84). This fraction can be determined by three methods under steady state conditions. The first method is by CO_2 ratio which expresses the specific activity of CO_2 produced after infusing [1-^{14}C] propionate relative to the specific activity of CO_2 produced after infusing [2-^{14}C] propionate. Similarly, the second method is by using the specific activity of blood glucose formed from the two types of labeled propionate. The third method is more complicated and involves comparing the specific activity of carbons 2 and 3 versus carbon 1 in glutamate when [2-^{14}C] propionate is the substrate.

Amino Acids and Other Glucose Precursors

Both Bergman (7) and Leng (55), who recently reviewed ruminant gluconeogenesis, have discussed the role of amino acids. Bergman (7) pointed out that the contribution of amino acids to gluconeogenesis will vary widely, being influenced by the nutritional and physiological status of the animal.

Black et al. (18) pointed out that the transfer quotient suffers the same "crossover" problems already discussed with propionate contributions to glucose. Further, the flux of amino acids through the plasma pool and their specific activity at the actual site of glucose synthesis must be considered to obtain accurate values of the contribution of amino acids to gluconeogenesis. An additional consideration is that most studies are of amino acids infused into peripheral circulation.

One of the first studies on the glucogenicity of amino acids was that of Hunter and Millson (41), who estimated that about 12% of milk lactose was derived from amino acids in lactating cows. They hydrolyzed a ^{14}C protein and intravenously administered the resulting mixture of labeled amino acids. Black and his colleagues (16, 18) used single injections of [^{14}C] amino acids to follow amino acid conversion to glucose in both cows and goats. The time interval between injection of each [^{14}C] amino acid and maximum specific activity in plasma glucose varied widely with different amino acids. It seemed to them that amino acids released from protein were utilized in a manner that results in a prolonged availability of gluconeogenic precursors so the animal can form glucose over long intervals. This should ensure a more continuous supply of gluconeogenic substrates and, in turn, a more efficient utilization of precursors.

Single injections of individual amino acids would have the advantages that pool sizes, half-life, irreversible loss, and time to maximum glucose specific activity can be measured, but continuous infusions or primed continuous infusions would have the advantage that a more accurate measure of individual amino-acid contributions to glucose can be obtained.

Most studies of amino acid contributions to glucose carbon in ruminants have been with sheep. Egan and Black (28), however, studied glutamate metabolism in a lactating cow by giving single, intravenous injections of labeled glutamate. The time course of the isotope was followed in respired CO_2, plasma glutamate, plasma glucose, and milk constituents. Egan and Black concluded that the amount of carbon from the plasma glutamate pool utilized for gluconeogenesis exceeds that incorporated into milk protein. Thus, gluconeogenesis acquires a larger fraction of the glutamate carbon than does protein synthesis during milk formation. At least 8% of the glucose carbon had been derived from plasma glutamate.

A variety of individual radioactive amino acids have been administered intravenously to estimate their contribution to gluconeogenesis in sheep or goats (29, 39, 100). Egan et al. (29) looked at the metabolic fate of uniformly labeled glutamate, valine, and arginine injected as a pulse dose into the portal vein of lactating goats. They found that glutamate was the most glucogenic of the three amino acids because ^{14}C from glutamate appeared earliest and in largest amounts in plasma glucose. More of the radioactive carbon of glutamate appeared in lactose than in milk protein. There was preferential utilization of carbon from valine and arginine for protein synthesis. The single-injection experiments do not allow an accurate measure of the contribution to glucose by individual amino acids.

Both Wolff and Bergman (100) and Heitmann et al. (39) used infusions lasting several hours to estimate the contribution of individual amino acids to gluconeogenesis in sheep. Wolff and Bergman (100) infused U-^{14}C-labeled alanine, aspartate, glutamate, glycine, and serine into the vena cava for 6 h. Alanine contributed 5.5% to the glucose turnover, aspartate .6%, glutamate 3.4%, glycine .9%, and serine .7%, giving a total of 11%. The results were not corrected for crossover. They found that the net hepatic uptake of alanine could produce a maximum of 1.6 millimoles of glucose per h, while the ^{14}C estimate indicated that 1.23 millimoles of glucose were derived from plasma alanine in the whole body. If some conversion of alanine to glucose occurred in the kidney, the small difference would suggest only a small error from not correcting for crossover. Heitmann et al. (39) used [U-^{14}C] glutamate and [U-^{14}C] serine and measured their gluconeogenic contribution in mature wether sheep. Glutamate contributed 6.2% of the plasma glucose and serine contributed 1.7% so that the total from the two amino acids was about 8%. Again, these results were not corrected for crossover.

Nolan and Leng (65) used measurements of urea entry rate in sheep to calculate the upper limit of amino-acid deamination and gave 50 to 60 g of glucose as the maximum potential availability of amino acid carbon. Bergman and his colleagues used arteriovenous differences and blood flow to measure amounts of unlabeled amino acids taken up and glucose produced by the liver (101) and by the kidney (11). Other studies on various aspects of the gluconeogenic contribution of amino acids have been conducted (19, 37, 72, 73, 100).

Contributions of other glucogenic precursors, including lactate, pyruvate, and glycerol, have been studied, but the same basic techniques already described have been used. Only compounds that have a net absorption from the gut give a net contribution to body glucose. Thus, Cori cycle lactate, blood pyruvate, and much of the glycerol would have arisen originally from body glucose or glycogen with no net contribution to glucose. In most dietary conditions, little or no lactate is produced or absorbed as a result of rumen metabolism. Nevertheless, glycerol in times of fasting and lactate after times of exercise make necessary contributions to keeping the body in glucose homeostasis.

In Vitro Studies of Ruminant Gluconeogenesis

Enzymatic, cellular, tissue, and organ studies of ruminant gluconeogenesis in vitro can be extremely helpful by isolating systems and reducing the variables such that gluconeogenic processes and controls can be discerned and studied more easily. The purpose of in vitro studies is to understand in vivo processes, and valid in vitro studies must be confirmed by studies involving living animals. In vitro studies can lead to erroneous conclusions. One example is the in vitro studies incubating rumen mucosa with propionate and observing the formation of lactate as a major end product (68, 82, 92, 94). One logical conclusion was that there is a major in vivo conversion of propionate to lactate during rumen absorption. In vivo studies by Weigand et al. (93) showed, however, that there is no major conversion of propionate to lactate by rumen mucosa, and their studies were supported by the work of Weekes (90).

Enzymatic studies can be used to verify that enzymes are present in great enough activity to support postulated metabolic pathways or that effective amounts of an enzyme are too low to support a postulated pathway. Ballard et al. (4) summarized early enzymatic studies relating to both ruminant gluconeogenesis and lipogenesis and pointed out that ruminant gluconeogenesis is different from gluconeogenesis in nonruminants. The ruminant liver lacks glucokinase, which is the adaptive and most active glucose-phosphorylating enzyme in nonruminants. Phosphoenolpyruvate carboxykinase, which is highly adaptive to gluconeogenic needs in rats, shows little adaptation in ruminants. The lipogenic enzymes, ATP-citrate lyase and NADP-malate dehydrogenase, are negligible in ruminant liver and adipose tissue; thus, ruminants have almost no capacity to synthesize fat from glucose.

Young et al. (103) assayed activities of phosphoenolpyruvate carboxykinase, malic enzyme, citrate cleavage enzyme, and glucose 6-phosphate dehydrogenase in homogenates of rumen mucosa, liver, and adipose tissue of cattle. The rumen mucosa contained considerably more activity of malic enzyme than of phosphoenolpyruvate carboxykinase. Thus, any conversion of propionate to lactate by rumen mucosa would likely involve malic enzyme. Because malic enzyme normally is associated with lipogenesis, it seemed unlikely that there was a major conversion of propionate to lactate, and this was later confirmed by Weigand et al. (93). The studies of Young et al. (103) also indicated that gluconeogenic enzymes of cattle adapt little to various physiologic conditions, such as fasting for 8 days or feeding a concentrate diet for at least 3 mo before slaughter. These studies can be interpreted to indicate that there is a continuous requirement for gluconeogenesis in cattle with little need for fluctuations in gluconeogenic enzyme activities.

There are more examples in sheep than in cattle where enzymatic studies have increased our knowledge of ruminant gluconeogenesis. Weekes (89) found that propionyl CoA synthetase activity in sheep rumen mucosa increased with pregnancy. Mackie and Campbell (62) used pregnant and lactating ewes and found that total glucose-6-phosphatase fell during pregnancy and rose during lactation in proportion to liver hypertrophy. Martin et al. (63) observed that long-term adaptations of sheep to fasting are characterized by a decrease in gluconeogenic enzymes. Assaying liver and kidney enzymes in fasted sheep, Filsell et al. (36) found that the activities of glucose-6-phosphatase, fructose-1,6-diphosphatase, and pyruvate carboxylase increased in both tissues. No increase occurred in the activity of phosphoenolpyruvate carboxykinaee either in nonpregnant, nonlactating fasted sheep, or in fasted sheep in the late stages of pregnancy. Butler and Elliot (21) studied phosphoenolpyruvate carboxykinase and found that decreased feed intake and glucocorticoid administration both decreased the enzymatic activity. In a different approach, Mathias and Elliot (64) studied propionate metabolism in nuclear-free homogenates of liver biopsies from lactating cows and found that the ability to metabolize propionate decreased as the stress of lactation increased.

Isolated liver cells prepared by the method of Berry and Friend (14) now are being used rather extensively to study gluconeogenesis in laboratory animals. At least one laboratory (42) is using this general procedure to obtain intact hepatocytes from lamb liver. Effects of hormones, substrate concentrations, etc., would be studied easily in such whole-cell preparations and should give results more indicative of in vivo situations than results from enzymatic studies using homogenate fractions.

Tissue-slice studies are another convenient way to study gluconeogenesis. Some problems are involved, however. Unless biopsy samples of liver are taken, the animal must be sacrificed to get samples of liver. The animal must be sacrificed to obtain kidney samples, and sacrifice of adult cows, for example, is expensive. If a trocar and cannular are used to obtain puncture biopsies of liver, only a small amount is obtained. The puncture biopsy technique has the advantage that serial samples can be taken from a given animal. There is a risk, however, because a major blood vessel could be punctured. Larger biopsy samples can be obtained by opening the animal surgically and removing part of a liver lobe. The large samples allow more incubations at a given time, but there are numerous difficulties in repeated sampling. Sasaki et al. (74) used kidney-cortex slices from fed and fasted sheep and found that both propionate and glutamate were capable of making major contributions to glucose synthesis. Krebs and Yoshida (54) measured gluconeogenesis from several substrates in kidney cortex slices of cattle and sheep. Cow liver slices obtained at slaughter were used by Seto et al. (76), who observed that propionate was highly glucogenic and that glucose and phosphoenolpyruvate inhibited glucose formation and CO_2 production from propionate.

In regards to whole-organ studies, the most common have been liver perfusions with livers from rats or guinea pigs. There have been few examples of perfusions of ruminant livers (61). The technique could be helpful, but the large livers and large amounts of perfusion medium required for ruminant livers would present major obstacles.

Bergman and his colleagues have done experiments where whole organs were isolated effectively in vivo by placing catheters in the blood vessels entering and leaving the liver or kidney (47). They have used this technique to measure substrate uptake and glucose output from both liver (48) and kidney (50, 51) of sheep in various physiological conditions. The same general techniques should be applicable in cattle, but the hepatic vein is not as distinct in cattle as in sheep.

CURRENT UNDERSTANDING OF RUMINANT GLUCONEOGENESIS

Considerable progress has been made in our understanding of ruminant gluconeogenesis in recent years, but many more studies will be required before animal scientists, physiologists, and biochemists have a thorough knowledge of the process and how it is altered in various physiological situations. To date, much effort has been devoted to developing and refining techniques necessary to study fundamental aspects of gluconeogenesis. Studies today generally are more refined and sophisticated than they were a few years ago. There is a need to use existing techniques to obtain additional fundamental data, and there is a need to continue development and refinement of experimental techniques.

Careful scrutiny of present literature reveals a strong need to quantitate all aspects of glucose metabolism and gluconeogenesis in ruminants under standardized conditions, so that comprehensive data are available for precise physiological situations. Much more research has been done with sheep than with cattle. The techniques discussed in this manuscript all have not been applied to animals of the same size, fed the same diet at the same short intervals. Only when such closely controlled studies are done for all aspects of gluconeogenesis can we hope to have an in-depth understanding of the fundamental processes, how they are controlled, and how they change with changing dietary and physiological situations.

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